PROJECT
MANAGEMENT
DOCS

STATEMENT OF WORK (SOW)

LITPATH AI: SMART PATHFINDER FOR THESES AND DISSERTATIONS

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PROJECT MANAGEMENT DOCS

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INTRODUCTION/BACKGROUND

LitPath AI: Smart Pathfinder for Theses and Dissertations is an AI-powered, standalone web application developed for the DOST-STII Library. Operating independently from the existing Online Public Access Catalog (OPAC), the project addresses critical limitations found in the current system, such as exact-match keyword searches that often return irrelevant results and slow response times that hinder research efficiency. The goal is to implement a dedicated, AI-powered search engine that significantly improves search precision and the overall discovery of the library's collection of graduate-level academic materials.

SCOPE OF WORK

The scope of the LitPath AI project includes the delivery of the following tangible outputs:

In-Scope:

1. AI-Powered Search Engine: Development of a semantic search engine utilizing Retrieval-Augmented Generation (RAG) and Gemini AI for advanced query understanding and re-ranking.
2. Standalone Web Application: A fully functional, independent user interface that operates separately from the existing DOST-STII OPAC.
3. Client-Provided Database: An independent backend database populated exclusively with the sample thesis and dissertation papers provided directly by DOST-STII for the system's search repository.
4. Automated Citation Generator: A built-in tool that automatically formats citations for selected research materials in APA, MLA, Chicago Manual of Style, and IEEE formats.
5. Research Management & Feedback Tools: A bookmarking feature allowing users to save and organize their discovered research materials, alongside an integrated feedback tool to capture user ratings and comments.
6. Admin Analytics Dashboard: A dedicated dashboard providing DOST-STII library administrators with visibility into search trends, usage patterns, and user interactions.

Out-of-Scope:

1. Digitization: Full-text access to or the physical digitization of non-digitized library materials.
2. Collection Management: Direct management, deletion, or alteration of the library’s primary collection records or physical materials.

PERIOD OF PERFORMANCE

The project spans the academic year 2025-2026. Key development milestones began with initial sprint planning in November 2025, with the roadmap extending through three academic terms, targeting final production deployment and handover by June–July 2026. Project execution is constrained by a part-time academic schedule, limiting capacity to 10 hours per week per team member.

PLACE OF PERFORMANCE

- Development Phase: Work is conducted on the development team's local environments.
- Testing Phase: Internal and unit testing will be conducted on a Proxmox server environment.
- Deployment: The final standalone system will be hosted in the client's (DOST-STII) designated environment. Cloud hosting and AI/LLM API costs will be fully covered by DOST-STII.

WORK REQUIREMENTS

As part of the Website Redesign Project APC-HecTech will be responsible for performing tasks throughout various stages of this project. The following is a list of these tasks which will result in the successful completion of this project:

Kickoff:

- APC-HecTech will create and present detailed project plan including schedule, WBS, testing plan, implementation plan, training plan, and transition plan
- APC-HecTech will present project plan to DOST-STII/ScINET-PHIL Library for review and approval

Design Phase:

- Work with DOST-STII/ScINET-PHIL to gather requirements and establish metrics.
- Create site design based on collected requirements.
- Develop site design proposal for DOST-STII/ScINET-PHIL review and approval.
- Present written status at weekly meeting.

Build Phase:

- APC-HecTech will complete all coding for approved site design
- APC-HecTech will provide DOST-STII/ScINET-PHIL with a detailed testing plan

- APC-HecTech will include all content provided by DOST-STII/ScINET-PHIL on redesigned web site
- APC-HecTech will conduct testing in both their iLab as well as in a limited beta release
- APC-HecTech will resolve any coding and site issues identified in testing
- APC-HecTech will compile a testing report to present to SCG for review/approval
- Present written status at weekly meeting

Implementation Phase:

- APC-HecTech will implement the newly redesigned web site on DOST-STII/ScINET-PHIL servers
- APC-HecTech will begin providing 24x7 web site support at this point forward until the end of the period of performance
- Present written status at weekly meeting

Training Phase:

- APC-HecTech will provide training in accordance with approved training plan provided in the kickoff
- Present written status at weekly meeting

Project Handoff/Closure:

- APC-HecTech will provide DOST-STII/ScINET-PHIL with all documentation in accordance with the approved project plan
- APC-HecTech will present project closure report to DOST-STII/ScINET-PHIL for review and approval
- APC-HecTech will complete the project requirements checklist showing that all project tasks have been completed
- APC-HecTech will conclude 24x7 web support at 11:59pm on the final day of the period of performance
- Present written status at weekly meeting

SCHEDULE/MILESTONES

The following milestones mark the key checkpoints of the LitPath AI project:

Milestone	Target Date	Status
Project Scoping	April 2025	Completed
Information Gathering	April 2025	Completed
Completion of Project Deliverables & Documentation	June 2025	Completed

Design Thinking Process (Stages 1–5)	Aug 2025	Completed
Core UI Development (System UI, Citation Generator, Bookmark)	Sep – Oct 2025	Completed
AI Search Engine, Generate Overview, Feedback Mechanism & Admin Dashboard	Oct 2025 – Jan 2026	Completed
Feature Refinement & Client Feedback Integration	Feb – Mar 2026	Completed
Continued Development & Performance Optimization (incl. codebase restructuring)	Mar – Apr 2026	In Progress
Proxmox Server Deployment & Internal Testing	Apr 2026	In Progress
Bug Fixes & Usability Testing	May – Jun 2026	Not Started
Final Deployment	Jun – Jul 2026	Not Started

ACCEPTANCE CRITERIA

For the LitPath AI Project, the acceptance of all deliverables will reside with the Project Sponsors, Khasian Eunice Romulo and Jonathan Abalon, representing DOST-STII.

Once a project phase is completed, the HecTech team will provide their report for review. The Project Sponsors will either sign off on the approval for the next phase to begin or provide actionable feedback. Schedule changes exceeding a 10% threshold variance must also be explicitly approved by the Project Sponsors.

Final acceptance of the project requires that:

1. The project delivers a 100% working standalone application with all defined features (AI search, citation generator, bookmarking, admin dashboard) fully completed.
2. The system achieves at least 70% relevance accuracy for its top 10 results.
3. The system achieves an average user satisfaction score of at least 3.5 out of 5 from the test group.

Once all project tasks have been completed, the project will enter the handoff stage. The HecTech team will provide their final report and documentation package. The formal sign-off by the Project Sponsors will acknowledge acceptance of all project deliverables.

OTHER REQUIREMENTS

- Database Constraints: The project team will utilize an independent database populated with sample papers provided by DOST-STII. Direct access to the live DOST-STII production database is not required and will not be provided.
- Data Privacy: Compliance with the Data Privacy Act of 2012 (RA 10173) is required for administrative accounts; public search functions must remain anonymous.
- Security: All connections must be secured via HTTPS/SSL.
- Personnel & Leadership: Development is executed exclusively by the four-member HecTech team with no external outsourcing. Project leadership and management responsibilities are shared equally among the team members: Tracie Tomon, Joshua Sta Ana, Jenine Elaine Dulay, and Charijoy Cempron.

SPONSOR ACCEPTANCE

Approved and accepted by:

_____ Date: April 1, 2026

Khasian Eunice Romulo
Science Research Specialist II / Unit Head
(Digital Information and Processing Delivery Unit)